

73. Auto-Negotiation for backplane and copper cable assembly

73.1 Auto-Negotiation introduction

Change the second paragraph of 73.1 as follows:

The Auto-Negotiation function allows an Ethernet device to advertise characteristics and modes of operation it possesses to another device at the remote end of a Backplane Ethernet link and to detect corresponding operational modes and characteristics that the other device may be advertising.

73.3 Functional specifications

Change the last sentence of the third paragraph of 73.3 as follows:

Technology-Dependent PHYs are those supported by the Auto-Negotiation process (see Table 73–4 and Table 73A–1a).

Change 73.4 as follows:

73.4 Transmit function ~~requirements~~

The Transmit function provides the ability to transmit DME pages (see 73.5.2). The first DME pages exchanged by the local device and its link partner after Power-On, link restart, or renegotiation is referred to as the Base Page and contains the base Base Page link codeword defined in Figure 73–6 (see 73.6.2).

To exchange information beyond what is available in the Base Page, additional DME pages may be transmitted using the Next Page function (see 73.4.2).

73.4.1 Technology ability

Multiple technologies may be advertised by the Auto-Negotiation process simultaneously. The advertisement of technologies the device supports is transmitted in the Base Page (see 73.6.2) and a specific Next Page (see 73A.1a).

The local device may modify the link codeword (see 73.6) to disable an ability it possesses, but ~~will not transmit~~ it shall not advertise an ability it does not possess. This makes possible the distinction between local abilities and advertised abilities so that multi-ability devices may Auto-Negotiate to a mode lower in priority than the highest common ability.

It is the responsibility of the Arbitration function (see 73.7.4) to determine the common mode of operation shared by a link partner and to resolve multiple common modes.

NOTE—Previous editions of this standard prohibited simultaneous advertisement of PHYs that support operation over electrical backplanes with PHYs that support operation over copper cable assemblies.

73.4.2 Next Page function

The Next Page function uses the Auto-Negotiation arbitration mechanisms to allow for the exchange of information beyond what is available in the Base Page. DME pages exchanged after the Base Page are referred to as Next Pages. Next page exchange occurs after the Base Page has been exchanged if either end of the link segment has set the Next Page bit to logical one in the link codeword (see 73.6.1.2) indicating that it has at least one Next Page to send. There are two encodings of the Next Page link codeword that may be sent during Next Page exchanges, specifically: Message code (see 73.6.3) and Unformatted (see 73.6.4).

Multiple Next Pages, using appropriate link codewords (Message code and Unformatted), are transmitted to send extended messages. Each series of Next Pages begins with the transmission of a Message code Next Page. Each Message code Next Page has its own specific format and requirements for whether any Unformatted Next Pages will follow (see Annex 73A).

Any number of Next Pages may be sent in any order, with the exception that if a Message code 2 Next Page (see 73A.1a) is to be sent it shall be the first Next Page transmitted following the Base Page exchange.

It is recommended that the total number of Next Pages sent be kept small to minimize the link startup time.

Next page transmission ends when both ends of a link segment set their Next Page bits to logical zero, indicating that neither has anything additional to transmit. It is possible for one device to have more pages to transmit than the other device. Once a device has completed transmission of its Next Page information, it shall transmit Message code 1 Next Pages (Null message code, see 73A.1) with the NP bit set to logical zero in the Null message link codeword while its link partner continues to transmit valid Next Pages. A device capable of Auto-Negotiation shall recognize reception of Message Pages with Null message codes as the termination of Next Page information from its link partner.

73.4.3 Transmit Switch function

During Auto-Negotiation and prior to entry into the AN_GOOD_CHECK state, which is depicted in the arbitration state diagram (Figure 73–11), the Transmit Switch function shall connect only the DME page generator controlled by the transmit state diagram (Figure 119–14) to the MDI.

Upon entry into the AN_GOOD_CHECK state, the Transmit Switch function shall connect the transmit path from a single technology-dependent (highest common denominator) PHY to the MDI.

When a PHY is connected to the MDI through the Transmit Switch function, the signals at the MDI shall conform to all of the PHY specifications within 20 ms of the AN_GOOD_CHECK state entry.

73.5 DME transmission

73.5.1 DME electrical specifications

Change the second paragraph of 73.5.1 (as amended by IEEE Std 802.3ck-2022) as follows:

For any multi-lane PHY, DME pages shall be transmitted only on lane 0. The transmitters on other lanes should be disabled as specified in 71.6.7, 84.7.7, 85.7.7, 92.7.7, 93.7.7, 94.3.6.7, 136.8.7, 137.8.7, 162.8.7, ~~or~~ 163.8.7, 178.8.7, or 179.8.7.

Replace Table 73–1 as follows:

Table 73–1—DME electrical characteristics

Parameter	Min Value	Max Value Case 1 ^a	Max Value Case 2 ^b	Units
Transmit differential peak-to-peak output voltage	600	1200	1000	mV
Receive differential peak-to-peak input voltage	200	1200	1200	mV

^a When not advertising any of the PHY types defined in Clause 178 or Clause 179

^b When advertising any of the PHY types defined in Clause 178 or Clause 179

73.6 Link codeword encoding

Change 73.6 and renumber its subclauses (as amended by IEEE Std 802.3ck-2022 and IEEE Std 802.3df-2024) as follows:

The 48-bit Auto-Negotiation page takes on three different encodings referred to as link codewords. The three different encodings of link codewords are as follows:

- Base Page link codeword
- Message code Next Page link codeword
- Unformatted Next Page link codeword

The mappings of the Auto-Negotiation page data bits (D[47:0]) to specific fields, for each type of link codeword, are shown in Figure 73–6, Figure 73–7 and Figure 73–8. Within a link codeword, D0 shall be the first bit transmitted after the Manchester violation (see Figure 73–5).

All link codeword encodings contain the Ack and NP fields described in 73.6.1. The encoding for the Base Page is described in 73.6.2. The encodings for Message code and Unformatted Next Pages are described in 73.6.3 and 73.6.4 respectively.

The mapping of all possible technologies available for backplane and copper cable assembly by Clause 73 Auto-Negotiation are allocated over two link codeword encodings. The mappings are defined in Table 73–4 and Table 73A–1a and are transmitted in the Base Page (see 73.6.2) and Message code 2 Next Page link codewords (see 73.6.3 and 73A.1a).

73.6.1 Common link codeword fields

The Ack and NP fields are present in all link codeword encodings and are used to handshake the flow of the Auto-Negotiation process between the two partners.

73.6.1.1 Acknowledge

Acknowledge (Ack) is used by the Auto-Negotiation function to indicate that a device has recognized a link codeword from its link partner. The Acknowledge Bit is encoded in bit D14 of the link codeword. If no Next Page information is to be sent, this bit shall be set to logical one in the link codeword after the reception of at least three consecutive and consistent DME pages (ignoring the Acknowledge bit value). If Next Page

information is to be sent, this bit shall be set to logical one after the device has successfully received at least three consecutive and matching DME pages (ignoring the Acknowledge bit value), and will remain set until the Next Page information has been loaded into the AN XNP transmit register (registers 7.22, 7.23, 7.24). In order to save the current received link codeword, it has to be read from the AN LP XNP ability register (register 7.25, 7.26, 7.27) before the Next Page of transmit information is loaded into the AN XNP transmit register. After the COMPLETE ACKNOWLEDGE state has been entered the link codeword will be transmitted at least six times. Entry into the COMPLETE ACKNOWLEDGE state is depicted in the arbitration state diagram (Figure 73–11).

73.6.1.2 Next Page

Next Page (NP) is encoded in bit D15 of the link codeword. Support of Next Pages is mandatory. If the device does not have any Next Pages to send, the NP bit shall be set to logical zero. If a device wishes to engage in Next Page exchange, it shall set the NP bit to logical one. If a device has no Next Pages to send and its link partner has set the NP bit to logical one, it shall transmit Next Pages with Null message codes and the NP bit set to logical zero while its link partner transmits valid Next Pages. Next page exchanges will occur if either the device or its link partner sets the Next Page bit to logical one. The Next Page function is defined in 73.4.2.

73.6.2 Base Page link codeword

The base link codeword (Base Page) transmitted within a DME page shall convey the encoding shown in Figure 73–6. The Auto-Negotiation function supports additional pages using the Next Page function. Encoding for the link codeword(s) used in the Next Page exchange are defined in 73.7.7. In a DME page, D0 shall be the first bit transmitted.

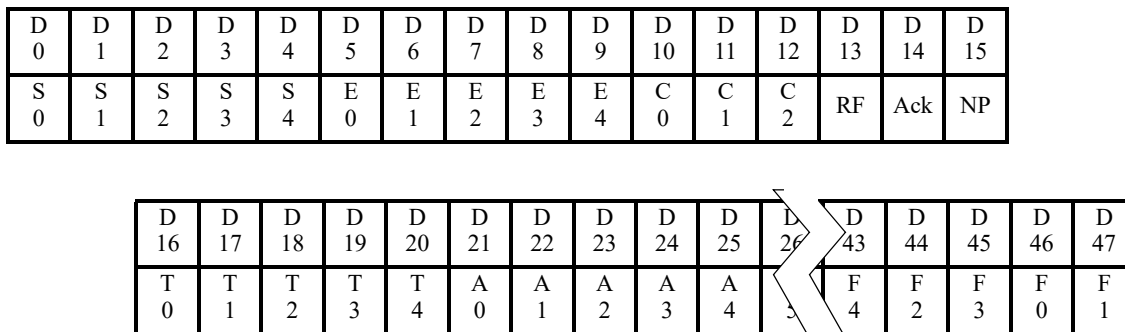


Figure 73–6— Link codeword Base Page link codeword

D[4:0] contains the Selector Field. D[9:5] contains the Echoed Nonce field. D[12:10] contains capability bits to advertise capabilities not related to the PHY. C[1:0] is used to advertise pause capability. The remaining capability bit C[2] is reserved. D[15:13] contains the RF, Ack, and NP bits. These bits shall function as specified in 28.2.1.2. D[20:16] contains the Transmitted Nonce field. D[42:21] contains the Technology Ability Field. D[47:43] contains FEC capability (see 73.6.5).

73.6.2.1 Selector Field

Selector Field (S[4:0]) is a five-bit wide field, encoding 32 possible messages. Selector Field encoding definitions are shown in Annex 28A. Combinations not specified are reserved for future use. Reserved combinations of the Selector Field shall not be transmitted.

The Selector Field for IEEE Std 802.3 is shown in Table 73–3.

Table 73–3—Selector Field Encoding

S4	S3	S2	S1	S0	Selector description
0	0	0	0	1	IEEE Std 802.3

73.6.2.2 Echoed Nonce Field

Echoed Nonce Field (E[4:0]) is a 5-bit wide field containing the nonce received from the link partner. When Acknowledge is set to logical zero, the bits in this field shall contain logical zeros. When Acknowledge is set to logical one, the bits in this field shall contain the value received in the Transmitted Nonce Field from the link partner.

73.6.2.3 Transmitted Nonce Field

Transmitted Nonce Field (T[4:0]) is a 5-bit wide field containing a random or pseudo-random number. A new value shall be generated for each entry to the Ability Detect state. The method of generating the nonce is left to the implementer. The transmitted nonce should have a uniform distribution in the range from 0 to $2^5 - 1$. The method used to generate the value should be designed to minimize correlation to the values generated by other devices.

73.6.2.4 Technology Ability Field

Technology Ability Field (A[21:0]) is a 22-bit wide field containing information indicating supported technologies specific to the selector field value when used with the Auto-Negotiation for backplane and copper cable assembly. These bits are mapped to individual technologies such that abilities are advertised in parallel for a single selector field value. The Technology Ability Field encoding for the IEEE 802.3 selector with Auto-Negotiation for backplane and copper cable assembly is described in Table 73–4.

~~Multiple technologies may be advertised in the link codeword. A device shall support the data service ability for a technology it advertises. It is the responsibility of the Arbitration function to determine the common mode of operation shared by a link partner and to resolve multiple common modes.~~

~~NOTE—Previous editions of this standard prohibited simultaneous advertisement of PHYs that support operation over electrical backplanes with PHYs that support operation over copper cable assemblies.~~

25GBASE-KR-S abilities are a subset of 25GBASE-KR abilities, and likewise 25GBASE-CR-S abilities are a subset of 25GBASE-CR abilities. To allow interoperation between 25GBASE-KR-S and 25GBASE-KR PHY types, and between 25GBASE-CR-S and 25GBASE-CR PHY types, a device that supports 25GBASE-KR or 25GBASE-CR should advertise both A9 and A10 ability bits during auto-negotiation.

The fields A[21:20] are reserved for future use. Reserved fields shall be sent as zero and ignored on receive.

Table 73–4—Technology Ability Field encoding

Bit	Technology
A0	1000BASE-KX
A1	10GBASE-KX4
A2	10GBASE-KR
A3	40GBASE-KR4
A4	40GBASE-CR4
A5	100GBASE-CR10
A6	100GBASE-KP4
A7	100GBASE-KR4
A8	100GBASE-CR4
A9	25GBASE-KR-S or 25GBASE-CR-S
A10	25GBASE-KR or 25GBASE-CR
A11	2.5GBASE-KX
A12	5GBASE-KR
A13	50GBASE-KR or 50GBASE-CR
A14	100GBASE-KR2 or 100GBASE-CR2
A15	200GBASE-KR4 or 200GBASE-CR4
A16	100GBASE-KR1 or 100GBASE-CR1
A17	200GBASE-KR2 or 200GBASE-CR2
A18	400GBASE-KR4 or 400GBASE-CR4
A19	800GBASE-KR8 or 800GBASE-CR8
A20 through A21	Reserved

73.6.2.5 FEC capability

FEC capability and request bits (F4, F2, F3, F0, F1) is are encoded in bits D43:D47 of the base link codeword. The FEC bits are used as follows:

- F0 is 10 Gb/s per lane FEC ability
- F1 is 10 Gb/s per lane FEC requested
- F2 is 25G RS-FEC requested
- F3 is 25G BASE-R FEC requested
- F4 is 100GBASE-P RS-FEC-Int requested

Bits F0 and F1 are used for 10 Gb/s per lane operation PHYs. F2 and F3 are used for resolving FEC operation for 25G PHYs. F4 is used by 100GBASE-P PHYs where RS-FEC-Int (see [Clause 161](#)) is an alternative to the default RS-FEC (see [Clause 91](#)).

73.6.2.5.1 FEC resolution for 100GBASE-P PHYs that include RS-FEC-Int

For 100GBASE-P PHYs that include RS-FEC-Int (see [Clause 161](#)) in addition to RS-FEC (see [Clause 91](#)), the F4 field is used to negotiate which FEC sublayer is to be used. If either PHY requests RS-FEC-Int operation then the RS-FEC-Int sublayer is enabled and the RS-FEC sublayer is disabled. Otherwise, the RS-FEC-Int sublayer is disabled and the RS-FEC sublayer is enabled.

73.6.2.5.2 FEC resolution for 25G PHYs

For 25G PHYs if neither PHY requests FEC operation in bits F2 or F3 then FEC is not enabled.

For 25GBASE-KR and 25GBASE-CR PHYs if either PHY requests RS-FEC then RS-FEC operation is enabled, otherwise if either PHY requests BASE-R FEC then BASE-R operation is enabled.

For 25GBASE-KR-S and 25GBASE-CR-S PHYs, if either PHY requests RS-FEC or BASE-R FEC then BASE-R operation is enabled. This is because 25GBASE-KR-S and 25GBASE-CR-S PHYs do not support RS-FEC operation.

73.6.2.5.3 FEC resolution for 10 Gb/s per lane PHYs

For 10 Gb/s per lane operation, when the FEC ability bit F0 is set to logical one, it indicates that the PHY has FEC ability (see [Clause 74](#)). When the FEC requested F1 bit is set to logical one, it indicates a request to enable FEC on the link.

Since the local device and the link partner may have set the FEC capability bits differently, the priority resolution function is used to enable FEC in the respective PHYs. The FEC function shall be enabled on the link if 10GBASE-KR, 40GBASE-KR4, 40GBASE-CR4, or 100GBASE-CR10 is the HCD technology (see 73.7.6), both devices advertise FEC ability on the F0 bits, and at least one device requests FEC on the F1 bits; otherwise FEC shall not be enabled.

~~The variable `an_rs_fec_int_negotiated_control` indicates that RS-FEC-Int operation has been negotiated. If the value is false, then RS-FEC-Int has not been negotiated. If the value is true, then RS-FEC-Int has been negotiated.~~

73.6.2.5.4 FEC control variables

The variable `an_baser_fec_control` indicates that BASE-R FEC operation has been negotiated. If the value is false, then BASE-R FEC has not been negotiated. If the value is true, then BASE-R FEC has been negotiated.

The variable `an_rs_fec_control` indicates that RS-FEC operation has been negotiated. If the value is false, then RS-FEC has not been negotiated. If the value is true, then RS-FEC has been negotiated.

~~The variable `an_rs_fec_int_negotiated_control` indicates that RS-FEC-Int operation has been negotiated. If the value is false, then RS-FEC-Int has not been negotiated. If the value is true, then RS-FEC-Int has been negotiated.~~

The mapping of these variables to MDIO register bits is defined in Table 73–6.

If `mr_autoneg_enable` (see 73.10.1) is false, the FEC function is controlled by implementation-dependent means.

73.6.2.6 Pause Ability

Pause (C0:C1) is encoded in bits D11:D10 ~~of the base link codeword~~. The two-bit Pause is encoded as follows:

- a) C0 is the same as PAUSE as defined in [Annex 28B](#)
- b) C1 is the same as ASM_DIR as defined in [Annex 28B](#)

The Pause encoding is defined in [Clause 28B.2](#), [Table 28B–2](#). The PAUSE bit indicates that the device is capable of providing the symmetric PAUSE functions as defined in Annex 31B. The ASM_DIR bit indicates that asymmetric PAUSE is supported. The value of the PAUSE bit when the ASM_DIR bit is set indicates the direction the PAUSE frames are supported for flow across the link. Asymmetric PAUSE configuration results in independent enabling of the PAUSE receive and PAUSE transmit functions as defined by Annex 31B. See [28B.3](#) regarding PAUSE configuration resolution.

73.6.2.7 Remote Fault

Remote Fault (RF) is encoded in bit D13 of the base link codeword. The default value is logical zero. The Remote Fault bit provides a standard transport mechanism for the transmission of simple fault information. When the RF bit in the AN advertisement register (register 7.16.13) is set to logical one, the RF bit in the transmitted base link codeword is set to logical one. When the RF bit in the received base Base Page link codeword is set to logical one, the Remote Fault bit in the AN LP Base Page ability register (register 7.19.13) will be set to logical one, if the management function is present.

73.6.3 Message code Next Page link codeword encoding

The Message code Next Page transmitted within a DME page shall convey the encoding shown in Figure 73–7.

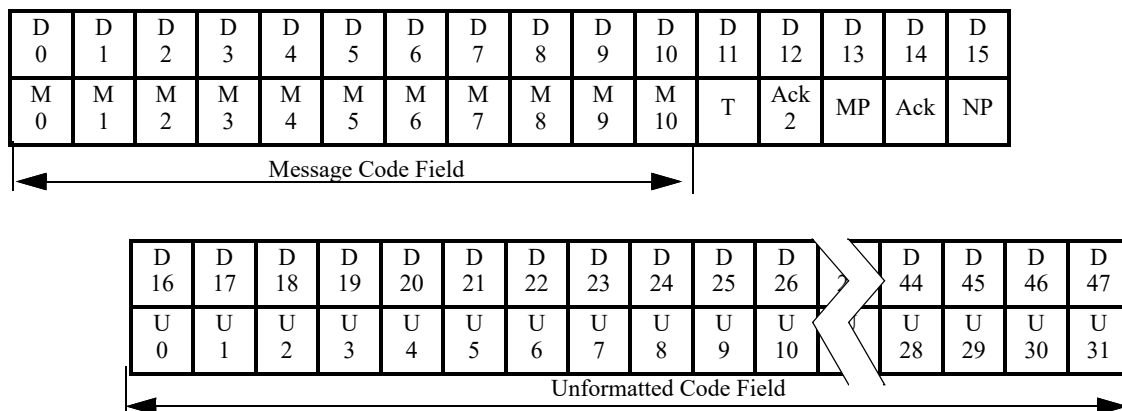


Figure 73–7— Message code Next Page link codeword

D[10:0] contains the Message Code Field, D[15:11] contains the T, Ack2, MP, Ack and NP bits. D[47:16] contains the Unformatted Code Field.

73.6.3.1 Message Code Field

Message Code Field (M[10:0]) is an eleven bit wide field, encoding 2048 possible messages. Message Code Field definitions are shown in Annex 73A. Combinations not specified are reserved for future use. Reserved combinations of the Message Code Field shall not be transmitted.

73.6.3.2 Toggle

Identical to that specified in [28.2.3.4.7](#).

73.6.3.3 Acknowledge 2

Identical to that specified in 28.2.3.4.6.

73.6.3.4 Message page

Identical to that specified in 28.2.3.4.5.

73.6.3.5 Unformatted code field

The Unformatted code field (U31:0) is a 32-bit generic container. The definition for each bit depends upon the Message code Field of the link codeword. See Annex 73A for details.

73.6.4 Unformatted Next Page link codeword encoding

The Unformatted Next Page transmitted within a DME page shall convey the encoding shown in Figure 73–8.

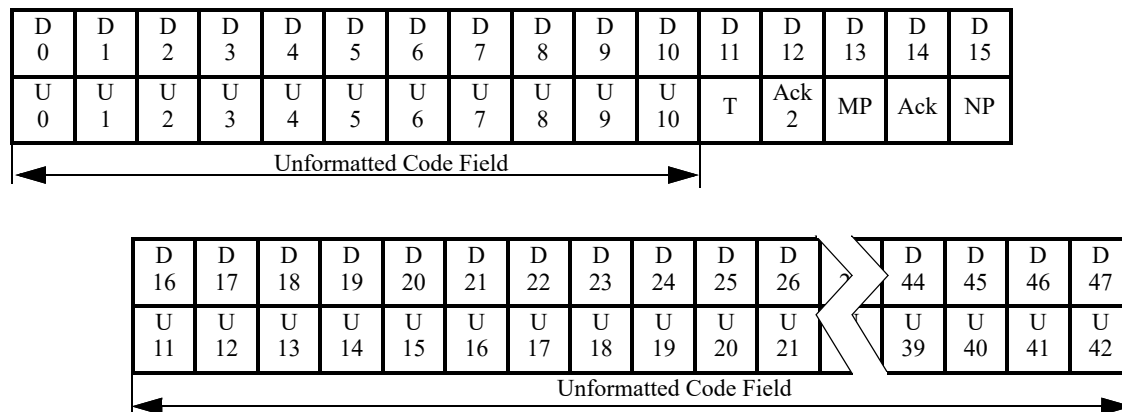


Figure 73–8—Unformatted Next Page link codeword

D[10:0] and D[47:16] contain the Unformatted Code Field, D[15:11] contains the T, Ack2, MP, Ack and NP bits.

73.6.4.1 Toggle

Identical to that specified in 28.2.3.4.7.

73.6.4.2 Acknowledge 2

Identical to that specified in 28.2.3.4.6.

73.6.4.3 Message page

Identical to that specified in 28.2.3.4.5.

73.6.4.4 Unformatted code field

The Unformatted code field (U42:0) is a 43-bit generic container. The definition for each bit depends upon the Message code Next Page that preceded the Unformatted Next Page. See Annex 73A for details.

73.6.8 Acknowledge

~~Acknowledge (Ack) is used by the Auto-Negotiation function to indicate that a device has successfully received its link partner's link codeword. The Acknowledge Bit is encoded in bit D14 of link codeword. If no Next Page information is to be sent, this bit shall be set to logical one in the link codeword after the reception of at least three consecutive and consistent DME pages (ignoring the Acknowledge bit value). If Next Page information is to be sent, this bit shall be set to logical one after the device has successfully received at least three consecutive and matching DME pages (ignoring the Acknowledge bit value), and will remain set until the Next Page information has been loaded into the AN_XNP transmit register (registers 7.22, 7.23, 7.24). In order to save the current received link codeword, it has to be read from the AN_LP_XNP ability register (register 7.25, 7.26, 7.27) before the Next Page of transmit information is loaded into the AN_XNP transmit register. After the COMPLETE ACKNOWLEDGE state has been entered, the link codeword will be transmitted at least six times.~~

73.6.9 Next Page

~~Next Page (NP) is encoded in bit D15 of link codeword. Support of Next Pages is mandatory. If the device does not have any Next Pages to send, the NP bit shall be set to logical zero. If a device wishes to engage in Next Page exchange, it shall set the NP bit to logical one. If a device has no Next Pages to send and its link partner has set the NP bit to logical one, it shall transmit Next Pages with Null message codes and the NP bit set to logical zero while its link partner transmits valid Next Pages. Next page exchanges will occur if either the device or its link partner sets the Next Page bit to logical one. The Next Page function is defined in 73.7.7.~~

73.6.10 Transmit Switch function

~~During Auto-Negotiation and prior to entry into the AN_GOOD_CHECK state, the Transmit Switch function shall connect only the DME page generator controlled by the Transmit State Diagram to the MDI.~~

~~Upon entry into the AN_GOOD_CHECK state, the Transmit Switch function shall connect the transmit path from a single technology dependent (highest common denominator) PHY to the MDI.~~

~~When a PHY is connected to the MDI through the Transmit Switch function, the signals at the MDI shall conform to all of the PHY's specifications within 20 ms.~~

Change 73.7 title as follows:

73.7 Receive function requirements

73.7.6 Priority Resolution function

Change the first paragraph of 73.7.6 as follows:

Since a local device and a link partner may have multiple common abilities, a mechanism to resolve which mode to configure is required. The mechanism used by Auto-Negotiation is a Priority Resolution function that predefines the hierarchy of supported technologies. The single PHY enabled to connect to the MDI by Auto-Negotiation shall be the technology corresponding to the bit in the Technology Ability Field (see 73.6.2.4) and the Extended Technology Ability Field (see 73A.1a) that is common to the local device and link partner that has the highest priority as defined in Table 73–5 (listed from highest priority to lowest priority). The Priority Resolution function shall only select technologies from the Extended Technology Ability Field when a Message code 2 Next Page is both transmitted and received.

Change the second sentence of the second paragraph as follows:

If the local device receives a Technology Ability Field or Extended Technology Ability Field with a bit set that is reserved, the local device shall ignore that bit for priority resolution.

Change Table 73–5 (as inserted by IEEE Std 802.3df-2024) as follows (some unchanged rows not shown):

Table 73–5—Priority Resolution

Technology	Capability
<u>1.6TBASE-KR8 or 1.6TBASE-CR8</u>	<u>1.6 Tb/s 8 lane, highest priority</u>
<u>800GBASE-KR4 or 800GBASE-CR4</u>	<u>800 Gb/s 4 lane</u>
800GBASE-KR8 or 800GBASE-CR8	800 Gb/s 8 lane, highest priority
<u>400GBASE-KR2 or 400GBASE-CR2</u>	<u>400 Gb/s 2 lane</u>
400GBASE-KR4 or 400GBASE-CR4	400 Gb/s 4 lane
<u>200GBASE-KR1 or 200GBASE-CR1</u>	<u>200 Gb/s 1 lane</u>
200GBASE-KR2 or 200GBASE-CR2	200 Gb/s 2 lane
200GBASE-KR4 or 200GBASE-CR4	200 Gb/s 4 lane
100GBASE-KR1 or 100GBASE-CR1	100 Gb/s 1 lane
...	

Delete subclause “73.7.7 Next Page Function”, as the contents of this section are now included within 73.4.

73.8 Management register requirements

Change Table 73–6 (as amended by IEEE Std 802.3ck-2022) as follows (some unchanged rows not shown):

Table 73–6—Backplane Ethernet Auto-Negotiation variable to MDIO register mapping

Variable	Description
mr_adv_ability[48:1]	{7.18.15:0, 7.17.15:0, 7.16.15:0} AN advertisement registers
mr_adv_extended_ability[32:1]	<u>{7.52.15:0, 7.53.15:0} extended technology advertisement registers</u>
mr_autoneg_complete	7.1.5 Auto-Negotiation Complete
mr_autoneg_enable	7.0.12 Auto-Negotiation Enable
mr_lp_adv_ability[48:1]	For Base Page: {7.21.15:0, 7.20.15:0, 7.19.15:0} AN LP Base Page ability registers For Next Page(s): {7.27.15:0, 7.26.15:0, 7.25.15:0} AN LP XNP ability registers
mr_lp_adv_extended_ability[32:1]	<u>{7.54.15:0, 7.55.15:0} LP extended technology ability registers</u>
...	

73.9 Technology-Dependent interface

73.9.1 AN_LINK.indication

Change 73.9.1.1 as follows:

73.9.1.1 Semantics of the service primitive

AN_LINK.indication(link_status)

The link_status parameter shall assume one of ~~two~~ three values: OK, ~~IN_PROGRESS~~, or FAIL, indicating ~~whether the underlying receive channel is intact and enabled (OK) or not intact (FAIL)~~ the status of the PCS alignment as defined in the PCS clause.

73.10 State diagrams and variable definitions

73.10.1 State diagram variables

Change the variable list in 73.10.1 (as amended by IEEE Std 802.3ck-2022 and IEEE Std 802.3df-2024) as follows (some unchanged list entries not shown):

200GR2;	represents the 200GBASE-KR2 or 200GBASE-CR2 PMD.
<u>200GR1;</u>	<u>represents the 200GBASE-KR1 or 200GBASE-CR1 PMD.</u>
400GR4;	represents the 400GBASE-KR4 or 400GBASE-CR4 PMD.
<u>400GR2;</u>	<u>represents the 400GBASE-KR2 or 400GBASE-CR2 PMD.</u>
800GR8;	represents the 800GBASE-KR8 or 800GBASE-CR8 PMD.
<u>800GR4;</u>	<u>represents the 800GBASE-KR4 or 800GBASE-CR4 PMD.</u>
<u>1.6TR8;</u>	<u>represents the 1.6TBASE-KR8 or 1.6TBASE-CR8 PMD.</u>

Insert new variable after mr_adv_ability variable as follows:

mr_adv_extended_ability[32:1]

A 32-bit array that contains the unformatted code field of transmitted message code 2 next pages.
How this status variable is updated is implementation specific and beyond the scope of this standard.

For each element within the array:

Values: zero; data bit is logical zero.
one; data bit is logical one.

Insert new variable after mr_lp_adv_ability variable as follows:

mr_lp_adv_extended_ability[32:1]

A 32-bit array that contains the unformatted code field of received message code 2 next pages.
How this status variable is updated is implementation specific and beyond the scope of this standard.

For each element within the array:

Values: zero; data bit is logical zero.
one; data bit is logical one.

Change the entry for single_link_ready in 73.10.1 (as amended by IEEE Std 802.3ck-2022 and IEEE Std 802.3df-2024) as follows:

single_link_ready

Status indicating that an_receive_idle = true and only one of the following indications is being received:

- 1) link_status_[1GKX] = OK
- 2) link_status_[2.5GKX] = OK
- 3) link_status_[5GKR] = OK
- 4) link_status_[10GKX4] = OK
- 5) link_status_[10GKR] = OK
- 6) link_status_[25GR] = OK
- 7) link_status_[40GKR4] = OK
- 8) link_status_[40GCR4] = OK
- 9) link_status_[50GR] = OK
- 10) link_status_[100GCR10] = OK
- 11) link_status_[100GKP4] = OK
- 12) link_status_[100GKR4] = OK
- 13) link_status_[100GCR4] = OK
- 14) link_status_[100GR2] = OK
- 15) link_status_[100GR1] = OK
- 16) link_status_[200GR4] = OK
- 17) link_status_[200GR2] = OK
- 18) link_status_[200GR1] = OK
- ~~19~~18) link_status_[400GR4] = OK
- 20) link_status_[400GR2] = OK
- ~~21~~19) link_status_[800GR8] = OK
- 22) link_status_[800GR4] = OK
- 23) link_status_[1.6TR8] = OK

73.10.2 State diagram timers

Change Table 73–7 (as amended by IEEE Std 802.3ck-2022 and IEEE Std 802.3df-2024) as follows:

Table 73–7—Timer min/max value summary

Parameter	Min	Value and tolerance	Max	Units
...				
link_fail_inhibit_timer (when the link is 100GBASE-KR1, 100GBASE-CR1, 200GBASE-KR1, 200GBASE-CR1, 200GBASE-KR2, 200GBASE-CR2, 400GBASE-KR2, 400GBASE-CR2, 400GBASE-KR4, 400GBASE-CR4, 800GBASE-KR4, 800GBASE-CR4, 800GBASE-KR8, or 800GBASE-CR8, or 1.6TBASE-CR8)	12.3		12.4	s
...				

73.11 Protocol implementation conformance statement (PICS) proforma for Clause 73, Auto-Negotiation for backplane and copper cable assembly¹

73.11.4 PICS proforma tables for Auto-Negotiation for backplane and copper cable assembly

73.11.4.3 Link codeword encoding

Change 73.11.4.3 as follows:

Item	Feature	Subclause	Value/Comment	Status	Support
LE1	Link codeword encoding	73.6 <u>73.6.2</u>	As shown in Figure 73–6	M	Yes []
LE2	First bit transmitted	73.6	D0	M	Yes []
LE3	RF, ACK, NP bits	73.6	As specified in 28.2.1.1	M	Yes []
LE4	Reserved Selector Field	73.6.1 73.6.2.1	Shall not be transmitted	M	Yes []
LE5	Echoed Nonce Field with Acknowledge set to zero	73.6.2 <u>73.6.2.2</u>	Contains logical zeros	M	Yes []
LE6	Echoed Nonce Field with Acknowledge set to one	73.6.2 <u>73.6.2.2</u>	Values received in Transmitted Nonce Field from Link Partner	M	Yes []
LE7	Transmitted Nonce Field	73.6.3 <u>73.6.2.3</u>	New value generated for each entry into Ability Detect	M	Yes []

¹Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

Item	Feature	Subclause	Value/Comment	Status	Support
LE8	Support of multiple technologies	73.6.4 <u>73.4.1</u>	Shall support all technologies advertised	M	Yes []
LE9	Technology ability reserved fields	73.6.4 <u>73.6.2.4</u>	Sent as zero and ignored by the receiver	M	Yes []
LE10	FEC capability resolution	73.6.5 <u>73.6.2.5.3</u>	Resolve enabling of FEC capability	M	Yes []
LE11	Acknowledge with no Next Page	73.6.8 <u>73.6.1.1</u>	Set to 1 after three DME pages	M	Yes []
LE12	Acknowledge with Next Page	73.6.8 <u>73.6.1.1</u>	Set to 1 after three DME pages	M	Yes []
LE13	Device has no Next Pages to send	73.6.9 <u>73.6.1.2</u>	Next Page bit set to 0	M	Yes []
LE14	Device has Next Pages to send	73.6.9 <u>73.6.1.2</u>	Next Page bit set to 1	M	Yes []
LE15	Transmit switch function upon entry into the AN_GOOD_CHECK state	73.6.10 <u>73.4.3</u>	Connect the transmit path of HCD PHY to the MDI	M	Yes []
LE16	Transmit switch function during Auto-Negotiation and prior to entry into the AN_GOOD_CHECK state	73.6.10 <u>73.4.3</u>	Connect only DME page generator to MDI	M	Yes []
LE17	PHY connection to MDI	73.6.10 <u>73.4.3</u>	Signals at MDI conform to all PHY specifications within 20 ms	M	Yes []

73.11.4.4 Receive function requirements

Change 73.11.4.4 as follows (most unchanged rows not shown):

Item	Feature	Subclause	Value/Comment	Status	Support
...					
RF16	Priority resolution with no common technology	73.7.6	HCD takes on value of NULL and link_status=FAIL	M	Yes []
<u>RF17</u>	<u>Priority resolution with MsgPg2 technology</u>	<u>73.7.6</u>	<u>Ignores message code 2 next page technologies unless sent by both ends of the link</u>	<u>M</u>	<u>Yes []</u>

73.11.4.5 Next Page function

Change 73.11.4.5 as follows (most unchanged rows not shown):

Item	Feature	Subclause	Value/Comment	Status	Support
NP1	Message codes	73.7.7 <u>73.4.2</u>	Each series of Next Pages has Message code	M	Yes []
NP2	Next Page transmission while link partner not done	73.7.7 <u>73.4.2</u>	Device transmits Null Message code and sets NP bit to 0	M	Yes []
NP3	Reception of Null message codes	73.7.7 <u>73.4.2</u>	Recognized as end of link partner's Next Pages <u>termination of Next Page information from link partner</u>	M	Yes []
NP4	Next page encoding	73.7.7.1 <u>73.6.3</u> <u>73.6.4</u>	As shown in Figure 73–7 and Figure 73–8	M	Yes []
NP5	NP, Ack, MP, Ack2, T bits	73.7.7.1	As specified in 28.2.3.4	M	Yes []
NP6	MP bit for message Next Pages	73.7.7.1	Set to logical one	M	Yes []
NP7	Message Code Field in message Next Page	73.7.7.1	Encoded in D[10:0]	M	Yes []
NP8	Unformatted Code Field in message Next Page	73.7.7.1	Encoded in D[47:16]	M	Yes []
NP9	MP bit for Unformatted Next Pages	73.7.7.1	Set to logical zero	M	Yes []
NP10	Unformatted Code Field in Unformatted Next Pages	73.7.7.1	Encoded in D[15:0] and D[47:16]	M	Yes []
NP11	Continuation of Next Page exchange	73.7.7.2	Exchange continues until NP bit is zero on both devices	M	Yes []
NP12	Transmission of Null Message Code field	73.7.7.2 <u>73.6.1.2</u>	Sent if device has no other information to transmit	M	Yes []
<u>NP13</u>	<u>Next page order</u>	<u>73.4.2</u>	<u>Message code 2 next pages are transmitted first</u>	<u>M</u>	<u>Yes []</u>

73.11.4.9 Auto-Negotiation annexes

Change 73.11.4.9 as follows (most unchanged rows not shown):

Item	Feature	Subclause	Value/Comment	Status	Support
AN13	AN message code 10	<u>73A.4</u>	EEE technology message code	M	Yes []
<u>AN14</u>	<u>EH0:EH1 advertisement</u>	<u>73A.1a</u>	<u>PMD shall be compliant to advertised host class</u>	<u>M</u>	<u>Yes []</u>